Fixed-Polylogarithmic Natural-Density Descent for the Collatz Map

Idris Ali Shaik

Independent researcher

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Abstract

Let

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ (3n+1)/2, & n \equiv 1 \pmod 2, \end{cases} \qquad T_{\min}(n) = \min_{k \ge 0} T^k(n).$$

Put

$$A_{FP} = \frac{1}{1 - H_2(\log_3 2)} = 19.9822266839\ldots, \qquad c_* = \frac{2}{\log(4/3)} = 6.9521189935\ldots,$$

where H_2 is binary entropy. For every fixed $A > A_{FP}$, $c > c_*$, and $\beta > 0$, we prove that all but $O(X/(\log X)^{\varkappa})$ integers $n \le X$ possess a shortcut-Collatz iterate, before $c \log n$ steps, satisfying

$$T^k(n) \le C(\log n)^A, \qquad \max_{0 \le j \le k} T^j(n) \le n^{1+\beta}.$$

The constants $C, \varkappa > 0$ may depend on the displayed fixed parameters. The endpoint $A = A_{FP}$ is not asserted.

At the weaker target $\exp((\log n)^{1-\delta})$, the same architecture gives the sharper exceptional count

$$O_{\delta,c,\beta}(X \exp(-c_{\delta,c,\beta}(\log X)^{1-\delta}))$$

for every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$. Companion consequences retain every raw clock constant greater than $3/\log(4/3)$, and the earlier smooth graded clock reaches n^{α} before

$$\left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon\right) \log n$$

shortcut steps.

The new assembly repeatedly re-certifies first-passage landings rather than transporting a generated distribution. Strictly nested thresholds identify every generated landing with a direct first passage from the original shell. An exact reverse-product loss, rescaled at the final threshold, reduces the shell failure profile to

$$e^{-cM} + M(L+1)(e^{-cL} + 2^{-L}).$$

An entropy-sharp maximal barrier and a two-regime choice of contraction parameters then separate the near-optimal clock from the terminal polylogarithmic exponent.

1. Introduction and main results

We work throughout with the shortcut Collatz map T displayed in the abstract. All unqualified logarithms are natural, and $\log_2$ denotes the base-two logarithm. Here $\mathbb{N} = \{1, 2, 3, \dots\}$. Put

$$a_0 = \frac{\log_2 3}{2}, \qquad \varrho = \frac{\sqrt{3}}{2}. \tag{1.1}$$

The proof below is organized around first entrances below nested shell-scaled targets. It does not estimate a fixed-time endpoint distribution. Every transport estimate is pointwise in the landing target, and every later certification failure is pulled back directly to the original dyadic shell.

This is a standalone argument. No fixed-time endpoint transport theorem and no generated-target equidistribution statement is a proof dependency. The separate Lean package now formalizes the optimized chain and its literal referee-facing fixed-polylogarithmic theorem. Its full build, dependency and trust audits, and the manuscript render audit are checked separately; none is a premise of the paper proof.

Define binary entropy by

$$H_2(p) = -p\log_2 p - (1-p)\log_2(1-p),$$

and put

$$p_* = \log_3 2, \qquad A_{FP} = \frac{1}{1 - H_2(p_*)}, \qquad c_* = \frac{2}{\log(4/3)}. \tag{1.2}$$

Theorem 1.1 (optimized fixed-polylogarithmic descent)

For every fixed

$$A > A_{FP}, \qquad c > c_*, \qquad \beta > 0, \tag{1.3}$$

there are constants $C, \varkappa, X_0 > 0$ with the following property. For every $X \geq X_0$, let $E_{A,c,\beta,C}(X)$ be the set of integers $1 \leq n \leq X$ for which no integer $0 \leq k < c\log n$ simultaneously satisfies

$$T^k(n) \leq C(\log n)^A, \qquad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\beta}.$$

Then

$$\# E_{A,c,\beta,C}(X) \leq C\frac{X}{(\log X)^{\varkappa}}. \tag{1.4}$$

In particular, $T_{\min}(n) \leq C(\log n)^A$ on a set of natural density one. The endpoint $A = A_{FP}$ is not asserted.

Theorem 1.2 (endpoint-rate stretched-logarithmic descent)

For every fixed $0 < \delta < 1$, $c > c_*$, and $\beta > 0$, there are constants $C_{\delta,c,\beta}, c_{\delta,c,\beta}, X_{\delta,c,\beta} > 0$ with the following property. For every $X \geq X_{\delta,c,\beta}$, let $E_{\delta,c,\beta}(X)$ be the set of integers $1 \leq n \leq X$ for which no integer $0 \leq k < c\log n$ simultaneously satisfies

$$T^k(n) \leq \exp((\log n)^{1-\delta}), \qquad \max_{0\leq j\leq k} T^j(n) \leq n^{1+\beta}.$$

Then

$$\# E_{\delta,c,\beta}(X) \leq C_{\delta,c,\beta} X \exp(-c_{\delta,c,\beta}(\log X)^{1-\delta}). \tag{1.5}$$

The endpoint $\delta = 1$ is not asserted.

Corollary 1.3 (raw clock)

Put

$$\mathrm{Col}(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2. \end{cases}$$

For every fixed $A > A_{FP}$, $\beta > 0$, and

$$c_{\mathrm{raw}} > \frac{3}{\log(4/3)}, \tag{1.6}$$

there are constants $C, \varkappa, X_0 > 0$ such that all but $CX/(\log X)^{\varkappa}$ integers $n \leq X$, for $X \geq X_0$, possess an integer $\ell < c_{\mathrm{raw}} \log n$ for which

$$\mathrm{Col}^{\ell}(n) \leq C(\log n)^A, \qquad \max_{0\leq j\leq \ell} \mathrm{Col}^j(n) \leq n^{1+\beta}.$$

The analogous statement with the target in Theorem 1.2 and its exceptional rate also holds. Since $3/\log(4/3) = 10.42817849\ldots$, the explicit constant 10.44 remains admissible.

Corollary 1.4 (fixed powers and the smooth graded clock)

For every fixed $\alpha > 0$, the exceptional set for $T_{\min}(n) > n^{\alpha}$ is

$$O_{\alpha,\sigma}(X \exp(-c_{\alpha,\sigma}(\log X)^{\sigma})) \tag{1.7}$$

for every fixed $0 < \sigma < 1$. If $0 < \alpha < 1$, then for every fixed $\varepsilon > 0$ the same fixed-power target is reached on a natural-density-one set before

$$\left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon\right) \log n \tag{1.8}$$

shortcut steps.

Relation to previous almost-all results

The theorem should be read along several independent axes: density notion, target scale, clock, and exceptional rate. The closest comparisons are:

Result	Status	Density	Target reached for almost all starts	Clock retained in the stated result	Quantitative exception
Korec [3]	published	natural	n^{θ}, every fixed $\theta > a_0$	not part of the cited theorem	not used here
Inselmann [2]	preprint	natural	n^{ε}, every fixed $\varepsilon > 0$	$2 \log n / \log(4/3)$ shortcut steps	density convergence
Tao [5]	published	logarithmic	every $f(n) \to \infty$	no single global clock in the headline theorem	logarithmic-density estimate
Mazur [4]	manuscript with a pinned formal artifact	natural	every $f(n) \to \infty$	$< 436 \log n$ raw steps	fixed-target logarithmic rate
Allikvere [1]	preprint	natural	every $f(n) \to \infty$	$< 12 \log n$ raw steps	$O((\log N_0)^{-1/29} + X^{-1/2000})$ for a fixed target
This paper	standalone optimized first-passage argument	natural	$C(\log n)^{A}$, every fixed $A > A_{\mathrm{FP}}$; also $\exp((\log n)^{1-\delta})$ with a stronger rate	every shortcut constant $> c_*$; every raw constant $> 3/\log(4/3)$; graded clock (1.8) for fixed powers	$O(X/(\log X)^{\varkappa})$ at the poly-log target; $O(X e^{-c(\log X)^{1-\delta}})$ at the stretched-log target

The fixed-polylogarithmic target is smaller than every fixed power and every fixed stretched-logarithmic target of the form in Theorem 1.2. It remains weaker than an arbitrary diverging function. Thus the present theorem does not supersede the arbitrary-threshold conclusions of [1] or [4], and the clock scale already appears in [2]. The comparison is deliberately coordinatewise: target scale, density notion, exceptional rate, and clock are not collapsed into one ordering.

The proof has six load-bearing steps.

1. The parity-vector bijection turns a complete dyadic shell into the uniform Boolean cube.
2. An adjustable maximal Boolean barrier gives an entropy-sharp shell exceptional rate and a certified all-prefix orbit envelope.
3. First-passage reversal gives a loss-filtered tagged-fiber bound for every target subset of a landing shell.
4. Strictly decreasing threshold ranks identify each recursively generated landing with a direct first passage from the original source shell.
5. Rank-scaled reverse loss and restriction to the first failed certification produce the $M(L+1)$ terminal profile.
6. A two-regime parameter choice separates the leading clock from the low-rank entropy rate and yields A_{FP}.

2. Density, parity, and affine iterates

For $S \subseteq \mathbb{N}$, put

$$B_S(X) = \#\{1 \le n \le X : n \notin S\}.$$

We call S (C, D)-**dense** if

$$B_S(X) \le C X^{1-D} \qquad (X \ge 1), \tag{2.1}$$

where $C > 0$ and $0 < D \le 1$. Every such set has natural density one.

We use the following varying-rate shell summation.

Lemma 2.1 (varying-rate dyadic summation)

Let $0 < \sigma \le 1$, $A, \gamma > 0$, and suppose that, for all sufficiently large M,

$$\#(E \cap [2^M, 2^{M+1})) \le Ae^{-\gamma(M+4)^\sigma} 2^M. \tag{2.2}$$

Then there are $\gamma' > 0$ and X_0 such that

$$\#(E \cap [1, X]) \le (2A + 1) X e^{-\gamma' (\log X)^\sigma} \qquad (X \ge X_0). \tag{2.3}$$

Proof

Let $2^J \le X < 2^{J+1}$. The shells below $J/2$ contain fewer than $X^{1/2}$ integers. On the remaining shells,

$$M + 4 \ge \frac{\log X}{4 \log 2}.$$

Thus their total contribution is at most

$$2AX \exp\left(-\frac{\gamma}{(4\log 2)^\sigma} (\log X)^\sigma\right).$$

Choose

$$\gamma' = \min\left\{\frac{\gamma}{(4\log 2)^\sigma}, \frac{1}{2}\right\}.$$

For all sufficiently large X, one has $X^{1/2} \le X e^{-\gamma' (\log X)^\sigma}$. Adding the early and late contributions proves (2.3).
□

For $n \ge 1$, define

$$p_i(n) = 1_{\{T^i(n) \text{ odd}\}}, \qquad s_k(n) = \sum_{i=0}^{k-1} p_i(n). \tag{2.4}$$

The exact parity coding below is classical in the stopping-time literature; see, for example, Terras [6]. Its short proof is included because no external result is needed by the argument.

Proposition 2.2 (parity-vector bijection)

For every $M \ge 0$,

$$n \bmod 2^M \mapsto (p_0(n), \ldots, p_{M-1}(n)) \tag{2.5}$$

is a bijection from $Z/2^M Z$ to $\{0, 1\}^M$.

Proof

The relation

$$n \equiv n' \pmod{2^{k+1}} \implies T(n) \equiv T(n') \pmod{2^k}$$

shows inductively that the first M parity bits depend only on $n \bmod 2^M$. For the converse, induct on the word length. Suppose the first bit is e, and use the induction hypothesis to recover the residue $m = T(n) \bmod 2^k$ from the remaining k bits. If $e = 0$, the unique compatible residue is

$$n \equiv 2m \pmod{2^{k+1}}.$$

If $e = 1$, it is

$$n \equiv (2m - 1)3^{-1} \pmod{2^{k+1}},$$

where 3 is a unit modulo 2^{k+1}. The first residue is even, the second is odd, and in either case applying T recovers $m \bmod 2^k$. Thus every word has exactly one preimage residue. □

Proposition 2.3 (exact affine iterate)

For every $n \geq 1$ and $k \geq 0$,

$$T^k(n) = \frac{3^{s_k(n)}}{2^k} n + \sum_{i=0}^{k-1} \frac{p_i(n) 3^{s_k(n) - s_{i+1}(n)}}{2^{k-i}}. \tag{2.6}$$

Proof

The assertion is trivial at $k = 0$. Passing from k to $k + 1$, an even letter divides every existing term by two. An odd letter multiplies every existing term by $3/2$ and adds $1/2$. This is exactly the displayed update. □

3. A dense maximal-barrier set

For a parity word of length M, put

$$Y_k = 2s_k - k, \qquad H_M = \max_{0 \leq k \leq M} \left| s_k - \frac{k}{2} \right| = \frac{1}{2} \max_{0 \leq k \leq M} |Y_k|. \tag{3.1}$$

For $0 \leq t < 1/2$, put

$$I(t) = D\left(\frac{1}{2} + t \,\middle\|\, \frac{1}{2}\right) = \left(\frac{1}{2} + t\right) \log(1 + 2t) + \left(\frac{1}{2} - t\right) \log(1 - 2t). \tag{3.2}$$

Lemma 3.1 (entropy-sharp maximal Boolean-walk bound)

If $M \geq 1$ and $0 \leq h < M/2$, then

$$2^{-M} \# \{w \in \{0,1\}^M : H_M(w) > h\} \leq 2 \exp(-M I(h/M)). \tag{3.3}$$

In particular, the right side is at most $2 \exp(-2h^2/M)$.

Proof

Fix $\theta > 0$ and stop the Boolean tree at the first prefix u with $|Y(u)| > 2h$. For a frontier node of depth k, set

$$\Phi_\theta(u) = 2^{-k} \cosh(\theta Y(u)) (\cosh \theta)^{M-k}.$$

The identity

$$\cosh(\theta(y - 1)) + \cosh(\theta(y + 1)) = 2 \cosh(\theta y) \cosh \theta$$

shows that replacing an unexpanded node by its two children preserves the total potential. The frontier formed by first-crossing prefixes and noncrossing depth-M words therefore has total $(\cosh \theta)^M$. Each crossing prefix contributes at least $2^{-|u|} \cosh(2\theta h)$, whence

$$\Pr(H_M > h) \leq 2 \exp(M \log \cosh \theta - 2\theta h). \tag{3.4}$$

Take $u = 2h/M$ and $\theta = \operatorname{arctanh} u$. The elementary Legendre identity

$$u\theta - \log\cosh\theta = D\left(\frac{1+u}{2}\,\Big\|\,\frac{1}{2}\right) = I(h/M)$$

proves (3.3). Pinsker's inequality in this binary case, or the direct convexity estimate $I(t) \ge 2t^2$, gives the final assertion. □

Define

$$r_k(n) = \sum_{i=0}^{k-1} \frac{p_i(n)}{3^{s_{i+1}(n)}2^{k-i}}, \qquad d_k = s_k - \frac{k}{2}. \tag{3.5}$$

By Proposition 2.3,

$$T^k(n) = \varrho^k n 3^{d_k} + (r_k(n)3^{k/2})3^{d_k}. \tag{3.6}$$

Lemma 3.2 (uniform affine correction)

If $H_M \le h$, then for every $1 \le k \le M$,

$$r_k(n)3^{k/2} \le (2+\sqrt{3})(1-\varrho^k)3^h < (2+\sqrt{3})3^h. \tag{3.7}$$

Proof

For $0 \le i < k$, the barrier gives $s_{i+1} \ge (i+1)/2 - h$. Hence

$$r_k 3^{k/2} \le 3^h \sum_{i=0}^{k-1} \frac{3^{(k-i-1)/2}}{2^{k-i}} = \frac{3^h}{2}\sum_{j=0}^{k-1} \varrho^j,$$

which is (3.7). □

For $0 < \eta \le 1$, let W_η be the set of integers n such that

$$\varrho^k n^{1-\eta} \le T^k(n) \le \varrho^k n^{1+\eta} \qquad (0 \le k \le \lfloor \log_2 n \rfloor). \tag{3.8}$$

Define

$$b_{ent}(\eta) = I\left(\frac{\eta}{\log_2 3}\right). \tag{3.9}$$

Proposition 3.3 (entropy-sharp density of the maximal-barrier set)

Let $0 < \eta < 1 - a_0$. For every fixed $0 < c < b_{ent}(\eta)$, there is $K_{\eta,c} > 0$ such that

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \le K_{\eta,c}e^{-cM} \qquad (M \ge 0). \tag{3.10}$$

Consequently W_η is $(K'_{\eta,c}, c/\log 2)$-dense after increasing the fixed constant.

Proof

Put $L_3 = \log_2 3$. Since $c < b_{ent}(\eta)$, choose a fixed $0 < \lambda < 1$ with

$$c < I\left(\frac{\lambda\eta}{L_3}\right). \tag{3.11}$$

On I_M, take

$$h = \frac{\lambda\eta M}{L_3}. \tag{3.12}$$

Assume $H_M \le h$. Since $3^h = 2^{\lambda\eta M} \le n^{\eta}$, the multiplicative term in (3.6) is at least $\varrho^k n^{1-\eta}$ and at most

$$2^{-(1-\lambda)\eta M}\varrho^k n^{1+\eta}. \tag{3.13}$$

Lemma 3.2 bounds the additive term by

$$(2+\sqrt{3})3^{2h} = (2+\sqrt{3})2^{2\lambda\eta M}. \tag{3.14}$$

The unused part of the upper envelope is at least

$$\left(1 - 2^{-(1-\lambda)\eta M}\right)\varrho^M 2^{(1+\eta)M}. \tag{3.15}$$

Its binary exponent exceeds that of (3.14), because

$$a_0 + \eta - 2\lambda\eta \ge a_0 - \eta > 0. \tag{3.16}$$

Thus (3.8) holds for every $k \le M$ once M is sufficiently large.

Proposition 2.2 and Lemma 3.1 now give

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \le 2\exp\left[-M I\left(\frac{\lambda\eta}{L_3}\right)\right] \le 2e^{-cM}$$

on all sufficiently large shells. Enlarge $K_{\eta,c}$ to absorb the finite startup. Summing the resulting geometric shell series proves the global density assertion. □

4. First-passage linear transport

For real $Y > 1$, define

$$\tau_Y(n) = \min\{h \ge 0 : T^h(n) \le Y\} \tag{4.1}$$

when the set is nonempty.

Lemma 4.1 (first-passage band and reverse product)

Let $1 < Y < 2^M$, let $n \in I_M$, and suppose $\tau_Y(n) = h \ge 1$. Write $x_j = T^j(n)$, $y = x_h$, and let s be the number of odd values among $x_0, \dots, x_{h-1}$. Then

$$\frac{Y}{2} < y \le Y \tag{4.2}$$

and

$$n = \frac{2^h y}{3^s} \prod_{\substack{0 \le j < h \\ x_j \text{ odd}}} \left(1 - \frac{1}{2x_{j+1}}\right). \tag{4.3}$$

Consequently, whenever $h < 2Y$,

$$\left(1 - \frac{h}{2Y}\right)\frac{2^h y}{3^s} \le n \le \frac{2^h y}{3^s}. \tag{4.4}$$

Proof

The final crossing cannot be odd: otherwise $x_h = (3x_{h-1} + 1)/2 > x_{h-1} > Y$. Hence $x_{h-1} = 2y$, proving (4.2). Reversing an even step gives $x_j = 2x_{j+1}$, while reversing an odd step gives

$$x_j = \frac{2x_{j+1} - 1}{3} = \frac{2}{3}x_{j+1}\left(1 - \frac{1}{2x_{j+1}}\right).$$

Multiplication proves (4.3). Every odd factor occurs before the final crossing, so its following state is greater than Y. Therefore $\prod_i(1 - u_i) \ge 1 - \sum_i u_i$ gives a lower product bound $1 - s/(2Y) \ge 1 - h/(2Y)$; the upper bound is one. □

For a tagged landing cell put

$$F_{M,Y}(h,y) = \#\{n \in I_M : \tau_Y(n) = h,\ T^h(n) = y\}. \tag{4.5}$$

Lemma 4.2 (odd-count rigidity)

If $h/(2Y) \le 1/3$, every source in a nonempty tagged fiber (h,y) has the same odd count.

Proof

If two sources had odd counts $s_1 < s_2$, put $A_i = 2^h y/3^{s_i}$. Then $A_1 \ge 3A_2$, and (4.4) would give

$$\frac{n_1}{n_2} \ge 3\left(1 - \frac{h}{2Y}\right) \ge 2.$$

This is impossible for two integers in the half-open shell I_M, whose ratio is strictly smaller than two. □

Lemma 4.3 (tagged-fiber bound)

If $h/(2Y) \le 1/3$, then

$$F_{M,Y}(h,y) \le 1 + \frac{2h2^M}{2Y - h}. \tag{4.6}$$

In particular, if $1 \le h \le H$ and $H/(2Y) \le 1/3$, then

$$F_{M,Y}(h,y) \le \frac{5}{2}H\frac{2^M}{Y}. \tag{4.7}$$

The estimates remain valid after arbitrary source restriction.

Proof

Lemma 4.2 fixes one odd count s. Put $A = 2^h y/3^s$ and $\varepsilon = h/(2Y)$. By (4.4), every source lies in $[(1-\varepsilon)A, A]$. Nonemptiness and $n < 2^{M+1}$ give

$$A < \frac{2^{M+1}}{1 - \varepsilon}.$$

The interval contains at most $1 + \varepsilon A$ integers, proving (4.6). Under the second hypotheses, the nonconstant term is at most $(3/2)H\,2^M/Y$, while $1 \le H\,2^M/Y$, proving (4.7). □

Put $J_Y = (Y/2, Y] \cap \mathbb{N}$.

Proposition 4.4 (arbitrary-target linear transport)

Let $1 < Y < 2^M$, $H \geq 1$, and $H/(2Y) \leq 1/3$. Every $B \subseteq J_Y$ satisfies

$$\#\{n \in I_M : \tau_Y(n) \leq H,\ T^{\tau_Y(n)}(n) \in B\} \leq \frac{5}{2} H^2 \frac{2^M}{Y} |B|. \tag{4.8}$$

The same inequality holds after arbitrary source restriction.

Proof

Every landing belongs to J_Y by (4.2). Since $Y < 2^M$, the passage time is positive. Sum (4.7) over the at most $H|B|$ tagged cells (h, y). □

For the optimized assembly, write the reverse product in (4.3) as $\prod_{j<h}(1 - u_j)$, where

$$u_j(n) = \begin{cases} \dfrac{1}{2T^{j+1}(n)}, & T^j(n) \text{ odd}, \\ 0, & T^j(n) \text{ even}, \end{cases} \qquad E_Y(n) = Y \sum_{j=0}^{h-1} u_j(n). \tag{4.9}$$

The elementary product inequality gives

$$1 - \frac{E_Y(n)}{Y} \leq \prod_{j=0}^{h-1} (1 - u_j(n)) \leq 1. \tag{4.10}$$

Lemma 4.5 (loss-filtered tagged fibers)

Let $D \geq 0$, and suppose

$$\frac{D}{Y} \leq \frac{1}{3}. \tag{4.11}$$

For a fixed tag (h, y), the number of sources $n \in I_M$ satisfying

$$\tau_Y(n) = h, \qquad T^h(n) = y, \qquad E_Y(n) \leq D$$

is at most

$$1 + 3D\frac{2^M}{Y}. \tag{4.12}$$

Proof

Put $P(n) = \prod_{j<h}(1 - u_j(n))$. If two such sources had odd counts $s_1 < s_2$, put $A_i = 2^h y/3^{s_i}$. Then $A_1 \geq 3A_2$, while (4.3) and (4.10) give

$$n_1 = A_1 P(n_1) \geq 3A_2\left(1 - \frac{D}{Y}\right) \geq 2A_2 \geq 2n_2.$$

This is impossible in the half-open shell I_M. The argument uses only positive products and introduces no division by a reverse product. Fix the common odd count and put $A = 2^h y/3^s$. All sources lie in $[(1 - D/Y)A, A]$. Nonemptiness, (4.11), and $n < 2^{M+1}$ imply $A < (3/2)2^{M+1}$, so this interval has length less than

$$3D\frac{2^M}{Y}.$$

This proves (4.12). □

Proposition 4.6 (loss-filtered target transport)

Let $1 < Y < 2^M$. Under (4.11), every $B \subseteq J_Y$ satisfies the exact bound

$$\#\left\{ n \in I_M : \begin{matrix} \tau_Y(n) \le H, & T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \le D \end{matrix} \right\} \le H\left(1 + 3D\frac{2^M}{Y}\right)|B|. \tag{4.13}$$

Consequently,

$$\#\left\{ n \in I_M : \begin{matrix} \tau_Y(n) \le H, & T^{\tau_Y(n)}(n) \in B, \\ E_Y(n) \le D \end{matrix} \right\} \le H(1 + 3D)\frac{2^M}{Y}|B|. \tag{4.14}$$

The estimate remains valid after arbitrary source restriction.

Proof

Since $Y < 2^M$, every passage time is positive. Sum (4.12) over the at most $H|B|$ tags to obtain (4.13). The contribution $H|B|$ is at most $H(2^M/Y)|B|$, which gives (4.14). □

5. Nested first-passage re-certification

Fix

$$a_0 < r < 1, \qquad 0 < \eta < r - a_0, \qquad 0 < c_\eta < b_{\mathrm{ent}}(\eta). \tag{5.1}$$

Increase a fixed startup rank $M_0 = M_0(r, \eta, c_\eta)$ whenever needed below. If $x \in W_\eta \cap I_m$, put $q = \lfloor rm \rfloor$. For all $m \ge M_0$,

$$T^m(x) \le \varrho^m x^{1+\eta} < 2^{(a_0+\eta)m+1+\eta} \le 2^q. \tag{5.2}$$

Thus $\tau_{2^q}(x) \le m$, and (3.8) bounds the whole block by $x^{1+\eta}$.

Fix integers $M \ge L \ge M_0$, and start with $n_0 = n \in I_M$. Put

$$m_i = \lfloor \log_2 n_i \rfloor, \qquad t_0 = 0.$$

If $m_i < L$, stop successfully. If $m_i \ge L$ but $n_i \notin W_\eta$, stop with a certification failure. Otherwise define

$$q_i = \lfloor rm_i \rfloor, \quad h_i = \tau_{2^{q_i}}(n_i), \quad n_{i+1} = T^{h_i}(n_i), \quad t_{i+1} = t_i + h_i. \tag{5.3}$$

The first-passage band gives

$$2^{q_i - 1} < n_{i+1} \le 2^{q_i}, \qquad m_{i+1} \in \{q_i - 1, q_i\}. \tag{5.4}$$

Since $q_i \le m_i - 1$,

$$m_{i+1} \le q_i \le rm_i, \qquad q_{i+1} \le q_i - 1 \tag{5.5}$$

whenever the next block is executed. Consequently

$$t_i \le \sum_{j<i} m_j \le M \sum_{j<i} r^j < \frac{M}{1-r}. \tag{5.6}$$

Lemma 5.1 (nested direct first passage)

For every executed block i,

$$t_{i+1} = \tau_{2^{q_i}}(n_0), \qquad n_{i+1} = T^{\tau_{2^{q_i}}(n_0)}(n_0). \tag{5.7}$$

Proof

The assertion is the definition when $i = 0$. Inductively, every state before t_i exceeds the preceding threshold and hence the strictly smaller 2^{q_i}. At time t_i, one has $n_i \geq 2^{m_i} > 2^{q_i}$. The local block stays above 2^{q_i} until its endpoint. Thus $t_i + h_i$ is exactly the first crossing of the new threshold by the original orbit. □

For $q \geq 1$, put

$$J_q = (2^{q-1}, 2^q] \cap \mathbb{N}, \qquad B_q = W_\eta^c \cap J_q. \tag{5.8}$$

The interval J_q lies in I_{q-1} apart from its upper endpoint. Proposition 3.3 therefore gives

$$\frac{|B_q|}{2^q} \leq C_{\eta,c_\eta}(e^{-c_\eta q} + 2^{-q}). \tag{5.9}$$

Lemma 5.2 (rank-scaled reverse loss)

If blocks $0, \ldots, i$ are executed before the first certification failure, then the direct first passage in (5.7) satisfies

$$E_{2^{q_i}}(n) < \frac{q_i + 2}{r}. \tag{5.10}$$

Proof

In block j, every state following an odd step is strictly larger than 2^{q_j}; the final crossing is even. Therefore each reverse-loss increment is less than 2^{-q_j-1}. Since the block has at most m_j steps, rescaling all blocks to the final threshold gives

$$E_{2^{q_i}}(n) < \frac{1}{2} \sum_{j \leq i} m_j 2^{q_i - q_j}.$$

Now $m_j < (q_j + 1)/r$, and the distinct integer thresholds q_j are all at least q_i. Hence

$$E_{2^{q_i}}(n) < \frac{1}{2r} \sum_{d \geq 0} (q_i + d + 1)2^{-d} = \frac{q_i + 2}{r}.$$

□

Let $\mathrm{Fail}_{M,L}$ denote the sources whose chain reaches a certification failure before it reaches rank below L.

Theorem 5.3 (optimized terminal profile)

Under (5.1), there is a fixed startup rank and a constant $C > 0$ such that, for all $M \geq L$ above that startup,

$$\boxed{\frac{\#\,\mathrm{Fail}_{M,L}}{2^M} \leq C\,[e^{-c_\eta M} + M(L + 1)(e^{-c_\eta L} + 2^{-L})].} \tag{5.11}$$

Every source outside this set has an integer $k < M/(1 - r)$ such that

$$T^k(n) < 2^L, \qquad \max_{0 \leq j \leq k} T^j(n) \leq n^{1+\eta}. \tag{5.12}$$

Proof

An initial failure has proportion $O(e^{-c_\eta M})$. Otherwise let the first failed certification be a generated landing in B_q. Every earlier block and the block producing that landing are certified, so Lemma 5.2 applies; the bad landing itself is not assumed certified. Lemma 5.1 makes it a direct first passage from I_M, at time at most $M/(1-r)+1$.

Increase the startup rank until

$$\frac{q+2}{r2^q} \le \frac{1}{3} \tag{5.13}$$

for every $q \ge L$. Proposition 4.6, with $D = (q+2)/r$, bounds the proportion of sources whose first failed landing has rank q by

$$CM(q+1)(e^{-c_\eta q} + 2^{-q}). \tag{5.14}$$

The two elementary tail estimates

$$\sum_{q \ge L} (q+1)e^{-c_\eta q} \ll (L+1)e^{-c_\eta L}, \qquad \sum_{q \ge L} (q+1)2^{-q} \ll (L+1)2^{-L}$$

prove (5.11).

Outside the failure set, integer ranks strictly decrease until a state of rank below L is reached. Equations (5.2), (5.6), and the monotonic decrease of block endpoints prove (5.12). □

6. Two-regime optimization

The near-minimal clock requires a contraction parameter close to a_0, which forces its envelope tolerance to be small. The terminal exponent is improved by using independent parameters at low ranks.

Choose

$$a_0 < r_{hi} < 1, \quad 0 < \eta_{hi} < r_{hi} - a_0, \quad 0 < c_{hi} < b_{ent}(\eta_{hi}), \tag{6.1}$$

and

$$a_0 < r_{lo} < 1, \quad 0 < \eta_{lo} < r_{lo} - a_0, \quad 0 < c_{lo} < b_{ent}(\eta_{lo}). \tag{6.2}$$

For an outer shell I_M, put

$$S_M = \lceil (\log(M+2))^2 \rceil. \tag{6.3}$$

Use the high triple when the current rank is at least S_M, the low triple between L_M and $S_M - 1$, and stop below L_M.

The proof of Lemma 5.1 is unchanged: if r_i is the active parameter, then $q_i = \lfloor r_i m_i \rfloor \le m_i - 1$, so

$$q_{i+1} \le m_{i+1} - 1 \le q_i - 1. \tag{6.4}$$

With $r_* = \min\{r_{hi}, r_{lo}\}$, the proof of Lemma 5.2 gives

$$E_{2^{q_i}}(n) < \frac{q_i+2}{r_*}. \tag{6.5}$$

The total shortcut time is at most

$$H_M^{(2)} \le \frac{M}{1 - r_{hi}} + \frac{S_M}{1 - r_{lo}}. \tag{6.6}$$

At a landing threshold $q \ge S_M + 1$, both possible landing ranks use the high certification set. At $q \le S_M - 1$, both use the low set. At the single boundary $q = S_M$, the landing target is the union of the two certification complements. Put

$$c_{sw} = \min\{c_{hi}, c_{lo}\}. \tag{6.7}$$

Theorem 6.1 (two-regime terminal profile)

If $L_M < S_M$ and both are above the fixed startup, then

$$\boxed{\begin{aligned}\frac{\#\ \mathrm{Fail}^{(2)}_{M,L_M}}{2^M} \ll\ & e^{-c_{hi}M} \\ & + M(S_M + 1)(e^{-c_{sw}S_M} + 2^{-S_M}) \\ & + M(L_M + 1)(e^{-c_{lo}L_M} + 2^{-L_M}).\end{aligned}} \tag{6.8}$$

On the complement, there is an integer $k \le H_M^{(2)}$ such that $T^k(n) < 2^{L_M}$. Moreover, if $\eta_{hi} < \beta$, then

$$\max_{0 \le j \le k} T^j(n) \le n^{1+\beta} \tag{6.9}$$

for every sufficiently large retained source.

Proof

Apply the first-bad proof of Theorem 5.3 separately in the high, switch, and low threshold ranges. At the switch, (6.7) bounds both certification tails; this is why only one union target is needed. The first two lines of (6.8) are superpolynomially small because $S_M \asymp (\log M)^2$.

High-phase blocks stay below $n^{1+\eta_{hi}}$. Every low-phase block starts below 2^{S_M+1}, so its complete orbit is at most

$$2^{(1+\eta_{lo})(S_M+1)} = \exp(O((\log M)^2)) = n^{o(1)}.$$

This proves (6.9). □

Proof of Theorem 1.1

As $\eta \uparrow 1 - a_0$,

$$\frac{1}{2} + \frac{\eta}{\log_2 3} \uparrow \log_3 2 = p_*.$$

Since relative entropy is continuous and

$$D(p_* \| 1/2) = (1 - H_2(p_*)) \log 2, \tag{6.10}$$

we have

$$\sup_{\eta < 1 - a_0} b_{ent}(\eta) = (1 - H_2(p_*)) \log 2. \tag{6.11}$$

Fix the parameters in (1.3). Choose

$$a_0 < r_{hi} < 1 - \frac{1}{c \log 2}, \qquad 0 < \eta_{hi} < \min\{r_{hi} - a_0, \beta\}. \tag{6.12}$$

Because $A > A_{FP}$, choose $\eta_{lo} < 1 - a_0$ with

$$A > \frac{\log 2}{b_{ent}(\eta_{lo})}.$$

Then choose

$$a_0 + \eta_{lo} < r_{lo} < 1$$

and $c_{lo} < b_{ent}(\eta_{lo})$ sufficiently close to the endpoint that

$$A > \frac{\log 2}{c_{lo}}. \tag{6.13}$$

Choose any admissible c_{hi}, and set

$$L_M = \lceil A \log_2(M + 2) \rceil. \tag{6.14}$$

Then $L_M < S_M$ eventually. The last line of (6.8) is $O(M^{-\varkappa})$ for every fixed

$$0 < \varkappa < \min\left\{ A - 1, \frac{Ac_{lo}}{\log 2} - 1 \right\}, \tag{6.15}$$

after decreasing $\varkappa$ slightly to absorb $L_M + 1$. The other lines are superpolynomially small. Also

$$2^{L_M} \le 2(M + 2)^A \ll_A (\log n)^A \qquad (n \in I_M). \tag{6.16}$$

Finally, (6.6) and (6.12) give

$$H_M^{(2)} \le \frac{\log n}{(1 - r_{hi}) \log 2} + O((\log \log n)^2) < c \log n \tag{6.17}$$

eventually.

For $2^J \le X < 2^{J+1}$, shells below $J/2$ contain $O(X^{1/2})$ integers, while on the remaining shells $M^{-\varkappa} \ll J^{-\varkappa}$. Their dyadic sizes sum to $O(X)$. This proves (1.4). □

The value in (1.2) is a nonattained infimum over fixed parameters.

7. Quantitative companions

Proof of Theorem 1.2

Fix $0 < \delta < 1$, put $\alpha = 1 - \delta$, and choose a one-regime pair (r, η) satisfying (5.1), with

$$r < 1 - \frac{1}{c \log 2}, \qquad \eta < \beta. \tag{7.1}$$

Take

$$L_M = \left\lfloor \frac{(M \log 2)^\alpha}{\log 2} \right\rfloor. \tag{7.2}$$

The factor $M(L_M + 1)$ in (5.11) is absorbed by the exponential tail, so

$$\# \ \mathrm{Fail}_{M,L_M} \le C 2^M e^{-c' M^\alpha}. \tag{7.3}$$

For $n \in I_M$,

$$2^{L_M} \le \exp((M \log 2)^\alpha) \le \exp((\log n)^\alpha).$$

The clock and path ceiling follow from (5.12) and (7.1). Lemma 2.1, with $\sigma = \alpha$, proves (1.5). □

Raw-clock conversion

Let $s_h(x)$ count the odd states among $x, T(x), \dots, T^{h-1}(x)$. Direct induction gives

$$\mathrm{Col}^{h+s_h(x)}(x) = T^h(x). \tag{7.4}$$

If $x \in W_\eta$ and $h \le \lfloor \log_2 x \rfloor$, the positive main term in the affine iterate and the upper envelope in (3.8) give

$$s_h(x) \le \frac{h}{2} + \frac{\eta \log x}{\log 3}. \tag{7.5}$$

Thus a one-regime chain has raw time at most

$$\frac{1}{1-r}\left(\frac{3}{2\log 2} + \frac{\eta}{\log 3}\right) \log n + o(\log n). \tag{7.6}$$

For the two-regime chain, (7.6) applies to the high phase; the entire low phase has $O(S_M) = o(\log n)$ shortcut steps and at most twice as many raw steps. As $r \downarrow a_0$ and $\eta \downarrow 0$, the coefficient in (7.6) tends to

$$\frac{3}{\log(4/3)}.$$

The high parameters in (6.12) can therefore be chosen to satisfy any fixed shortcut and raw budgets strictly above their respective limits. To retain the literal raw-orbit ceiling in Corollary 1.3, run the argument with an internal exponent $0 < \beta' < \beta$. Every additional raw odd image is $3m + 1 = 2T(m)$, so $2n^{1+\beta'} \le n^{1+\beta}$ eventually. This proves Corollary 1.3. □

Fixed-power exceptional count

Fix $\alpha > 0$ and $0 < \sigma < 1$. Choose $0 < \delta < 1 - \sigma$. Then

$$\exp((\log n)^{1-\delta}) \le n^\alpha$$

eventually, while $1 - \delta > \sigma$. Theorem 1.2 therefore implies (1.7), after decreasing the positive exponential constant and absorbing the finite startup.

8. Companion graded fixed-power clock

For completeness, we retain the fixed-depth stopped-map pullback that gives the sharper graded clock. This companion is not a dependency of Theorems 1.1 or 1.2.

Fix r, η as in (5.1), put $Y_M = 2^{\lfloor rM \rfloor}$, enlarge W_η on a finite startup interval, and let $G(n)$ be the first passage below Y_M on the certified part and the identity otherwise. For $S \subseteq \mathbb{N}$, set

$$\mathrm{FPPull}_{r,\eta}(S) = \tilde{W}_\eta \cap G^{-1}(S). \tag{8.1}$$

Proposition 8.1 (dense-set pullback)

For every fixed $0 < \chi < r$, there are $K_{FP}, D_c > 0$ such that, whenever $0 < D \leq D_c$, (8.1) sends every (C, D)-dense set to a

$$(K_{FP}(C + 1)D^{-2}, \chi D)\text{-dense set}. \tag{8.2}$$

Proof

For a source shell I_M, the bad landing target has cardinality at most $C Y_M^{1-D}$. Proposition 4.4 gives at most $O(CM^2 2^M Y_M^{-D})$ such sources. Since

$$Y_M^{-D} \leq 2\,2^{-rDM}$$

and

$$(M + 1)^2 e^{-(r-\chi)DM \log 2} \ll_{r,\chi} D^{-2},$$

this is $O((C + 1)D^{-2} 2^{(1-\chi D)M})$. Proposition 3.3 controls the certification complement at a fixed positive exponent; choose D_c so that this exponent is at least χD_c. Finite startup and geometric shell summation give (8.2). □

Put $g = 1 - a_0$. The envelope also gives the height-sensitive block bound

$$\tau_{Y_M}(n) \leq \left\lceil \frac{(1 + \eta - r)M + 2 + \eta}{g} \right\rceil \tag{8.3}$$

for every sufficiently large certified source in I_M. Indeed, $\varrho^h = 2^{-gh}$, and the displayed ceiling makes $\varrho^h n^{1+\eta} \leq 2^{\lfloor rM \rfloor}$.

Fix $0 < \alpha < 1$ and $\varepsilon > 0$. Choose $0 < \alpha' < \alpha$ with

$$c_*(\alpha - \alpha') < \frac{\varepsilon}{3},$$

then choose a fixed R and

$$r = (\alpha')^{1/R} > a_0, \qquad r^R = \alpha'.$$

Finally choose $0 < \eta < r - a_0$ so that

$$c_*(1 - \alpha')\frac{\eta}{1 - r} < \frac{\varepsilon}{3}. \tag{8.4}$$

Repeated use of Proposition 8.1 at this fixed depth produces a power-dense, hence natural-density-one, set on which all R blocks are certified. If n_i are the block endpoints, then

$$n_i \leq K_* n^{r^i} \tag{8.5}$$

for a fixed K_*. Summing (8.3) gives

$$\begin{aligned} k_R &\le \frac{1+\eta-r}{(1-a_0)\log 2}\frac{1-r^R}{1-r}\log n + O_{\alpha,\varepsilon}(1) \\ &= c_*(1-\alpha')\left(1+\frac{\eta}{1-r}\right)\log n + O_{\alpha,\varepsilon}(1). \end{aligned} \tag{8.6}$$

The choices leave a strict margin for the fixed remainder. Also $n_R \le K_* n^{\alpha'} \le n^{\alpha}$ eventually. This proves the graded clock in (1.8). □

9. Scope

The argument proves an orbit-minimum statement on a set of natural density one. It does not prove descent for every starting value, exclude nontrivial cycles, or control the orbit after the selected witness. The endpoint $\delta = 1$ is a strict parameter limit and is not claimed. No finite computation is used as an all-depth premise.

The headline proof consumes only the parity-vector bijection, the maximal-barrier estimate, the first-passage reversal identity, the exact reverse-product loss, and nested direct re-certification. The dense-set pullback in Section 8 is used only for the companion graded clock. In particular, no fixed-time endpoint-fiber moment or generated-target equidistribution hypothesis is used.

Research and software disclosure

Generative-AI systems were used in proof exploration, adversarial auditing, finite diagnostics, formalization, and exposition. The author selected the theorem and proof architecture, reviewed the resulting artifacts, and accepts responsibility for the manuscript. Finite diagnostics are supporting evidence only and are not premises of any theorem.

The separate Lean package formalizes the structural chain used above, including loss-filtered transport, nested re-certification, the two-regime terminal profile, strict endpoint-parameter selection, the literal terminal witness, and its same-witness orbit ceiling. Its public `Main` theorem has the same strict ranges $A > A_{FP}$, $c > c_*$, and $\beta > 0$ as Theorem 1.1, and its `badCount` is taken directly over the integers lacking that displayed witness; no auxiliary good-set quantifier remains in the public statement. The Lean landing inequality is strict, and therefore slightly stronger than the manuscript's weak landing inequality. The full package build, declaration-level dependency report, placeholder scan, and public-root axiom audit pass; the reported logical dependencies are `propext`, `Classical.choice`, and `Quot.sound`. These artifacts are additional verification evidence, not manuscript premises.

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